SOFTWARE REQUIREMENTS SPECIFICATION

for

MathVerse AI

AI-Powered Gamified Mathematics Learning Platform

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1 Introduction

1.1 Purpose

Conventional mathematics practice systems give every student the same experience and usually report only a final score. A correct answer does not explain why a wrong answer occurred, how long the student struggled, whether hints were needed, or whether the student is improving over time. Teachers and parents therefore have limited visibility into how a student learns.

"MathVerse AI" is a web-based, AI-powered and gamified mathematics learning platform that addresses this gap. It turns practice into story-based missions in which students solve questions, earn rewards and receive contextual help from an AI tutor named *Gwen*. The platform records correctness, response time, hint usage and interaction signals, maintains a personalised student memory, and adapts the difficulty of future activities. This document specifies the requirements of version 1.0 of the system.

1.2 Intended Audience and Reading Suggestions

This SRS is intended for the following readers.

- **Developers** should read Sections 2, 3 and 4 in full.
- **Testers** should focus on Section 3 (functional requirements) and Section 4 (non-functional requirements).
- **Course instructor and reviewers** may begin with Section 1 and the diagrams in Section 2.
- **Project managers** should read Sections 1.3, 1.4, 2.6 and 3.1.

1.3 Project Scope

MathVerse AI creates a shared space for **Students**, **Teachers** and **Parents**.

A student registers and logs in, selects a mission, and solves questions that are chosen according to topic, mastery and difficulty. Each attempt is verified and scored, and the system records the correctness, response time and hint usage. When a student is stuck, Gwen provides syllabus-grounded hints and explanations using Retrieval-Augmented Generation (RAG). The student's short-term and long-term memory is updated after attempts, and an adaptive engine uses this to advance strong performance or trigger remediation after repeated errors. Students earn points and badges and can view their progress.

Teachers and parents log in to view learning reports and track the progress of students linked to them.

Figure 1 gives an overview of how the main entities interact.

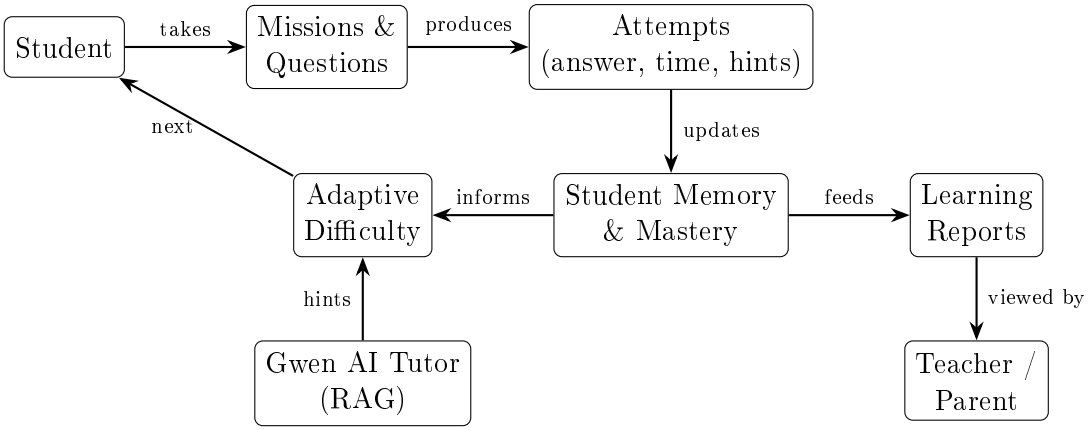


Figure 1: Entire work-flow of MathVerse AI

In Figure 1, the student takes missions; each mission produces attempts; attempts update the student memory; memory informs adaptive difficulty and feeds the learning reports that teachers and parents view. Gwen supports the student during an attempt.

1.4 Objectives

1. Create an engaging mathematics platform organised around missions, questions, objectives, rewards and progression.
2. Provide contextual hints, explanations and mistake analysis through the Gwen AI tutor.
3. Maintain short-term and long-term student memory for meaningful personalisation.
4. Use learning performance and behavioural signals to support adaptive difficulty.
5. Record attempts, correctness, response time and hint usage for analytics and evaluation.
6. Provide dashboards for students, teachers and parents.
7. Design the platform for scalable web deployment.

1.5 Stakeholders

Stakeholder	**Interest in the system**
Students	Primary users. Want engaging practice, helpful hints and visible progress.
Teachers	Want to see learning reports and identify students who need support.
Parents	Want visibility into their child's practice, strengths and weaknesses.
Development team	Design, implement and test the platform.
Course instructor (Miss Anushka)	Reviews the SRS and evaluates the project against the approved proposal.

Table 1: Stakeholders

1.6 Definitions and Abbreviations

Term	Meaning
Mission	A story-based set of mathematics questions on a topic and difficulty.
Mastery	A measure of how well a student understands a topic.
Gwen	The AI tutor that provides hints and explanations.
RAG	Retrieval-Augmented Generation: retrieving syllabus content before generating an answer.
LSTM	Long Short-Term Memory network used to analyse sequences of student behaviour.
Adaptive difficulty	Selecting the next question level from performance and behavioural signals.
API	Application Programming Interface.

Table 2: Definitions

2 Overall Description

2.1 Product Perspective

MathVerse AI is a new, self-contained web application. It replaces repetitive, fixed-difficulty practice with a personalised, mission-based experience. The main goal is to give each student suitable practice and to give teachers and parents meaningful insight beyond a final score.

The system follows a layered architecture (Figure 2): a frontend web application, a backend API, a persistent database, and an AI/analytics layer. The backend invokes the AI/analytics layer *asynchronously*, so a slow AI response never blocks the student's next question. Dashboards read from the Database Layer directly rather than passing through the AI layer.

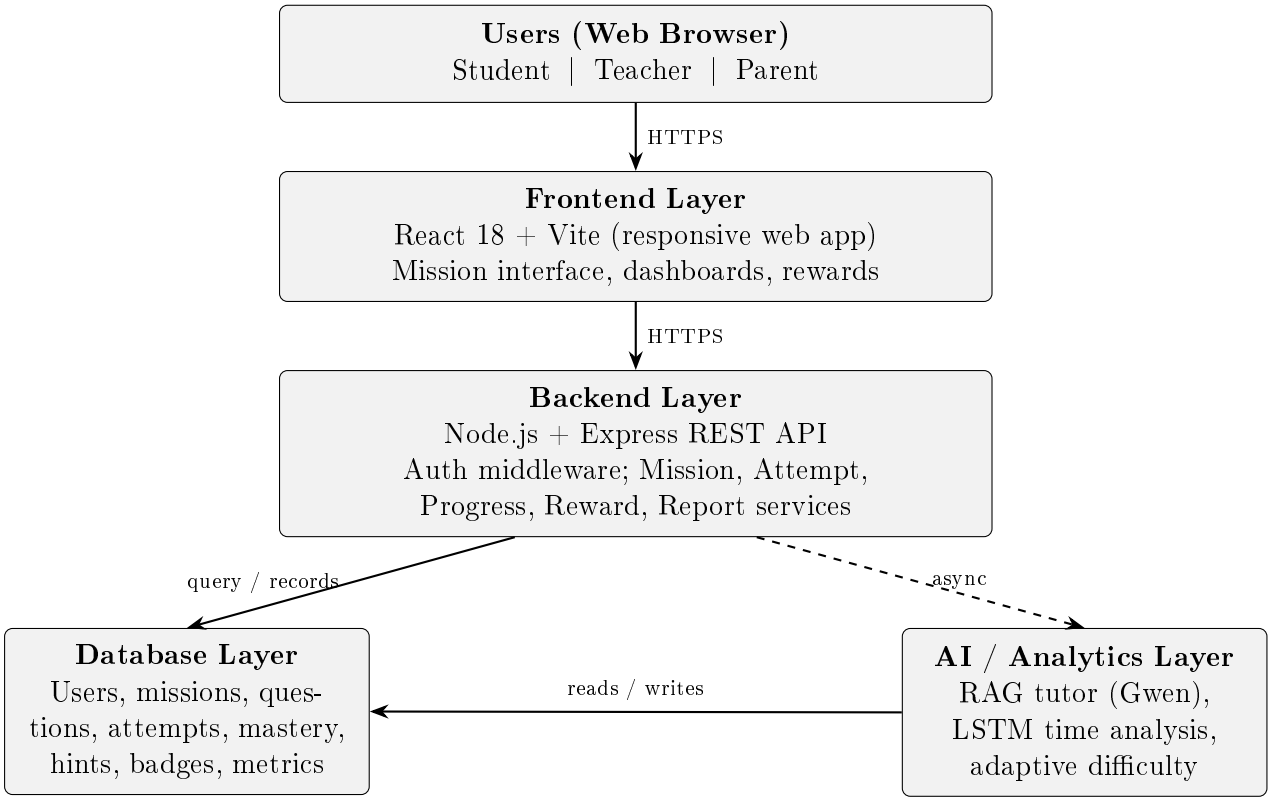


Figure 2: Layered system architecture of MathVerse AI

2.2 User Classes and Characteristics

MathVerse AI has three types of users.

User class	Characteristics and permitted actions
Student	Registers and logs in, selects missions, solves questions, asks Gwen for hints, earns rewards and views own progress.
Teacher	Logs in and views learning reports and progress of students.
Parent	Logs in and views learning reports and progress of their child.

Table 3: User classes

All users must be registered and authenticated. Each user can perform only the actions of their own role.

2.3 Product Functions

Before using the main functions, users must register. The system stores the following data (Figure 3).

- **User:** identifier, name, role, points, level.
- **Mission:** identifier, topic, difficulty.
- **Question:** identifier, mission, text, difficulty.
- **MissionAttempt:** identifier, user, score, time taken, hint used.
- **TopicMastery:** identifier, user, topic, mastery level.
- **HintHistory:** identifier, attempt, hint text, timestamp.
- **LearningReport:** identifier, user, total score, badges.
- **Badge:** identifier, criteria.

Student memory is divided into *short-term memory* (current mission, recent mistakes, current difficulty, recent hints and response behaviour) and *long-term memory* (topic mastery, strengths, weaknesses, repeated mistakes, learning patterns and historical performance).

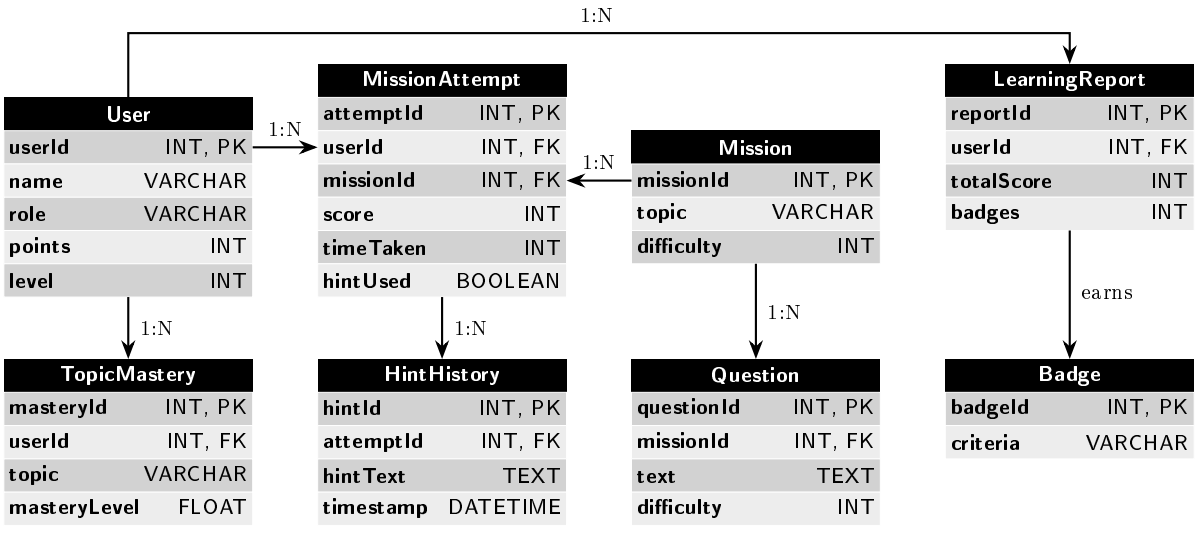


Figure 3: Database schema overview (entity relationships)

2.4 Operating Environment

The system is a web application accessed through a modern web browser on Windows, macOS or Linux, and on mobile devices through a responsive interface. The backend runs on Node.js and requires network access to the database and the AI/analytics services.

2.5 Design

2.5.1 Use Case Diagram

Students register or log in, play missions and track progress. Answer verification is included in mission play, and asking Gwen for a hint is optional. Teachers and parents view learning reports.

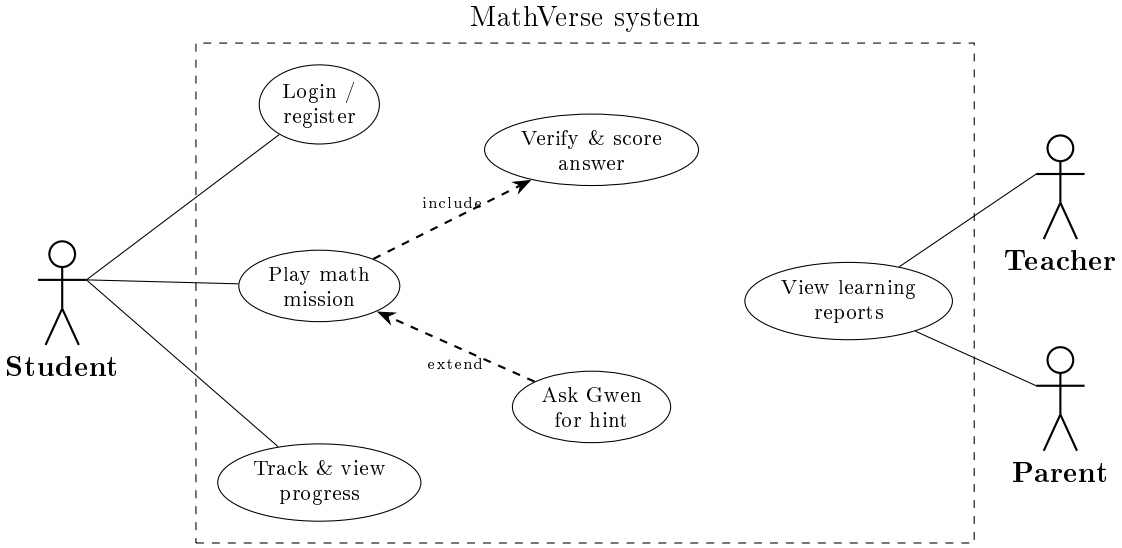


Figure 4: Use case diagram of MathVerse AI

2.5.2 Student Learning Journey

The student logs in and selects a mission; the system loads a question; the student solves and submits an answer; the system verifies, scores and updates mastery; AI analysis of behaviour and risk runs asynchronously; if more questions remain the loop continues, otherwise progress and badges are updated and a report is generated.

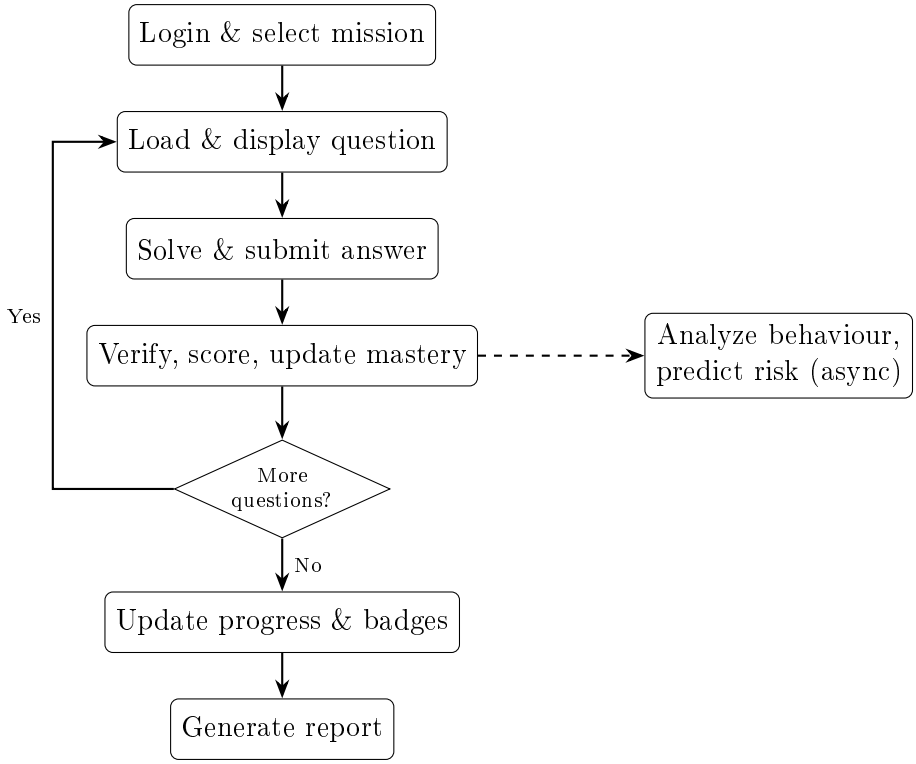


Figure 5: Activity diagram of the student learning journey

2.6 Design and Implementation Constraints

- The system must be developed as a web application using the technologies listed in Section 3.3.
- The project must be completed within the semester schedule by a team of four students.
- AI explanations must be grounded in curated, syllabus-aligned content.
- Training the LSTM model requires sufficient interaction data; until real data is available, evaluation may use sample or simulated data.
- Eye-tracking depends on external hardware/SDK and is outside the current build (see Section 5).
- Student data, particularly of minors, must be handled with privacy controls.

2.7 Assumptions and Dependencies

- Users have a modern browser and a stable internet connection.
- Curated syllabus content is available for the RAG tutor.
- The AI/analytics services and the database are reachable from the backend.
- Parents and teachers are linked to the correct students by the system.

3 System Features

3.1 Description and Priority

Features are listed from highest to lowest priority.

No.	Feature	Priority	Description
1	Authentication & roles	High	Registration, login and role-based access for all users.
2	Mission & question delivery	High	Missions and questions chosen by topic, mastery and difficulty.
3	Answer verification & attempt logging	High	Verify answers and record correctness, time and hints.
4	Student memory & mastery	High	Update short-term and long-term memory after attempts.
5	Gwen AI tutor (RAG)	High	Syllabus-grounded hints, explanations and mistake analysis.
6	Adaptive difficulty	Medium	Advance strong students or remediate after repeated errors.
7	Rewards & progression	Medium	Points, levels and badges.
8	Dashboards & reports	Medium	Student, teacher and parent views.
9	LSTM temporal analysis	Medium	Analyse sequences of response time, correctness and hints.
10	Eye-tracking signals	Low (planned)	Attention and re-reading signals; future scope.

Table 4: Key features and priorities

3.2 Functional Requirements

FR-1: Authentication and User Access

- FR-1.1: The system shall allow Students, Teachers and Parents to register and log in.

- FR-1.2: The system shall deny access to unregistered or unauthenticated users.
- FR-1.3: The system shall restrict each user to the actions permitted for their role.

FR-2: Mission and Question Workflow

- FR-2.1: The system shall allow a student to select a mission.
- FR-2.2: The system shall deliver questions based on topic, mastery and difficulty.
- FR-2.3: The system shall accept answer submissions and verify correctness.
- FR-2.4: The system shall record correctness, response time and hint usage for every attempt.

FR-3: Student Memory

- FR-3.1: The system shall maintain short-term memory (current mission, recent mistakes, current difficulty, recent hints).
- FR-3.2: The system shall maintain long-term memory (topic mastery, strengths, weaknesses, repeated mistakes, historical performance).
- FR-3.3: The system shall update mastery after each attempt.

FR-4: Gwen AI Tutor

- FR-4.1: The system shall allow a student to request a hint from Gwen during a question.
- FR-4.2: Gwen shall retrieve syllabus-aligned definitions, examples and formulas before generating an explanation.
- FR-4.3: The system shall store each hint in the hint history.

FR-5: Adaptive Difficulty and Temporal Analysis

- FR-5.1: The system shall analyse Question $\rightarrow$ Response Time $\rightarrow$ Correctness $\rightarrow$ Hint Usage $\rightarrow$ Next Question sequences.
- FR-5.2: The system shall increase difficulty after strong performance and trigger remediation after repeated errors.
- FR-5.3: AI analysis shall run asynchronously and shall not block the next question.

FR-6: Rewards and Progress

- FR-6.1: The system shall award points and update the student level.
- FR-6.2: The system shall unlock badges when their criteria are met.
- FR-6.3: The system shall show the student their rewards and progress.

FR-7: Dashboards and Reports

- FR-7.1: The system shall generate a learning report on mission completion.
- FR-7.2: Teachers and Parents shall be able to view learning reports and progress.
- FR-7.3: Dashboards shall read data from the Database Layer and shall not depend on the AI layer being available.

3.3 Technology Stack

- **Frontend:** React 18, Vite, JavaScript.
- **Backend:** Node.js, Express (REST API).
- **Database:** Persistent relational database (SQL, accessed through Prisma migrations).
- **AI/Analytics:** RAG pipeline for Gwen, LSTM model for temporal analysis, adaptive difficulty engine.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

- Normal pages and question loads should respond within about 2 seconds under typical load.
- AI processing shall run asynchronously so a slow AI response does not delay the next question.
- The system should use indexing, pagination and stateless services to support many concurrent users.

4.2 Security Requirements

- Only registered and authenticated users can enter the system.
- Role-based access control: a user can perform only the actions of their section.
- Passwords shall be stored as salted hashes, and all communication shall use HTTPS.
- Student data shall be accessible only to the student and their linked teacher and parent.

4.3 Privacy Requirements

- Behavioural data shall be used only for learning support and reporting.
- Behavioural signals are supporting evidence, not definitive judgements about a student.
- Any future eye-tracking shall include explicit privacy controls and conservative interpretation.

4.4 Software Quality Attributes

- **Usability:** responsive, game-like interface that is easy for students to use.
- **Reliability:** dashboards and core question flow remain available even if the AI layer is down.
- **Scalability:** layered, service-oriented design suitable for scalable web deployment.
- **Maintainability:** thin route $\rightarrow$ service layering and conceptual separation of AI services.
- **Correctness:** AI explanations must be factually correct and syllabus-relevant.

- **Testability:** functional, database, logical and UI tests, plus evaluation of accuracy improvement and reduction in repeated mistakes.

4.5 Business Rules

- MathVerse AI is used for mathematics practice and for tracking learning progress.
- Teachers and parents can view reports but cannot alter a student's attempts.
- Hints requested from Gwen are recorded and counted in the student's analytics.

5 Other Requirements

5.1 Maintenance

The system will need continued maintenance, refactoring and updates as curriculum and requirements change.

5.2 Planned and Future Scope

The following are part of the target design but are **not** part of the current build.

- Eye-tracking hardware SDK integration (with privacy controls).
- Email/SMS report delivery and push notifications.
- Third-party curriculum API integration.
- Expanded parent/teacher dashboards with intervention recommendations.
- Training and validation of the LSTM model and full RAG pipeline on curated content.